

ASYNCHRONOUS

Definition

"Asynchronous Javascript allows code to run without blocking other operations."

Purpose

Used for:

- APIs
- database requests
- authentication
- timers
- real-time applications

1. Callback Function

```
function fetchData(callback) {
```

```
  callback();
```

```
}
```

2. Callback Hell

Nested callbacks causing unreadable code.

3. Promise

Promise handles asynchronous operations.

4. States

- Pending
- Resolved
- Rejected

5. Promise Syntax

```
let promise = new Promise((resolve, reject) => {  
  resolve("Success");  
});
```

6. then() and catch()

Promise

- then(result => console.log(result))
- catch(error => console.log(error));

7. Async / Await

Modern way to handle

async code.

```
async function loadData() {  
  let data = await fetchData();  
}
```

8. Promise.all()

Runs multiple promises

together.

```
Promise.all([task1(), task2()]);
```

Real World Use

Async JavaScript is

- Used in:
- login System
 - fetching APIs
 - chat apps
 - payment gateways
 - live dashboards

Imp. Points: async/await improves readability

Promises solve callback hell & asynchronous code improves performance.